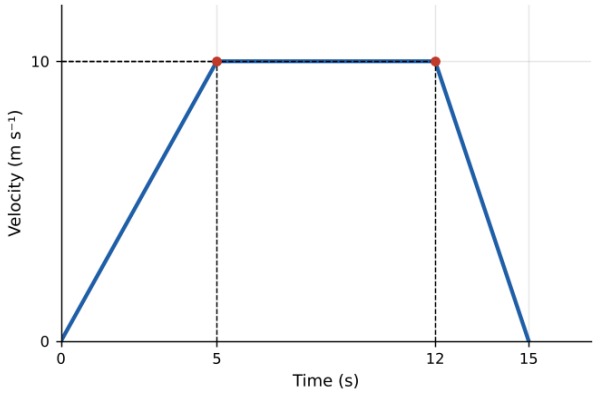
Science subjects

Physics

20 questions

Diagram questions

1. The velocity–time graph of a moving body is shown in the diagram. Calculate the total distance covered by the body in 15 s.

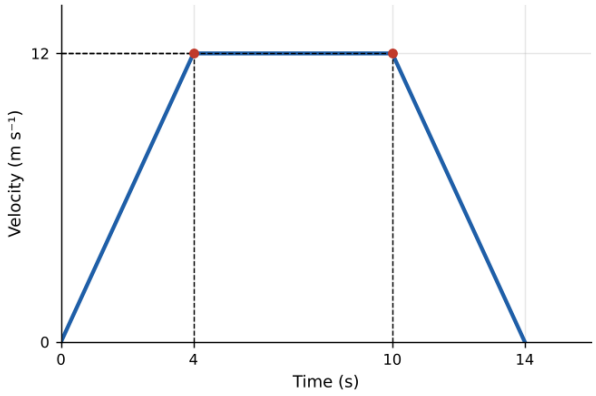


A. 75 m
B. 110 m
C. 120 m
D. 150 m

Answer: B

Distance = area under the velocity–time graph = ½×5×10 + (12−5)×10 + ½×3×10 = 25 + 70 + 15 = 110 m.

2. The graph shows the motion of a body along a straight line. Calculate the deceleration of the body during the last 4 seconds of its motion.

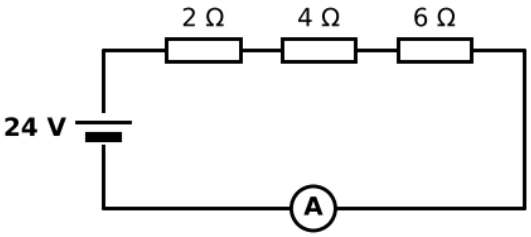


A. 1.2 m s⁻²
B. 4 m s⁻²
C. 12 m s⁻²
D. 3 m s⁻²

Answer: D

Deceleration = magnitude of the slope of the last section = (12 − 0)/(14 − 10) = 12/4 = 3 m s⁻².

3. In the circuit diagram shown, the battery has negligible internal resistance and the ammeter A has negligible resistance. What is the reading of the ammeter?

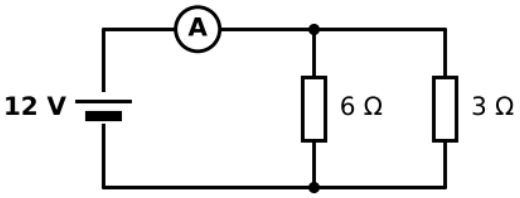


A. 2 A
B. 4 A
C. 6 A
D. 12 A

Answer: A

The resistors are in series: R = 2 + 4 + 6 = 12 Ω. Current I = V/R = 24/12 = 2 A.

4. In the circuit shown, the battery has negligible internal resistance. What is the reading of the ammeter A in the main line?

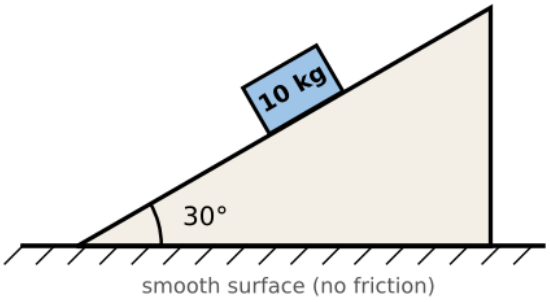


A. 1.3 A
B. 2 A
C. 6 A
D. 4 A

Answer: C

For the parallel resistors, 1/R = 1/6 + 1/3 = 1/2, so R = 2 Ω. The ammeter in the main line reads I = V/R = 12/2 = 6 A (2 A in the 6 Ω branch + 4 A in the 3 Ω branch).

5. A block of mass 10 kg slides down a smooth plane inclined at 30° to the horizontal, as shown. Calculate the acceleration of the block. [g = 10 m s⁻²]

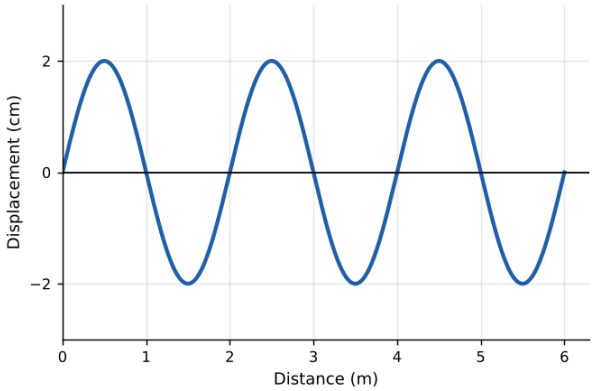


A. 10 m s⁻²
B. 5 m s⁻²
C. 8.7 m s⁻²
D. 50 m s⁻²

Answer: B

Along a smooth incline, a = g sin θ = 10 × sin 30° = 10 × 0.5 = 5 m s⁻². The mass cancels out, so it does not affect the acceleration.

6. The diagram shows a transverse wave travelling at a speed of 6 m s⁻¹. What is the frequency of the wave?

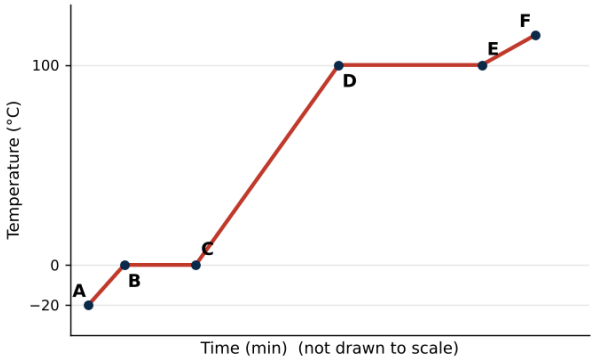


A. 3 Hz
B. 0.33 Hz
C. 4 Hz
D. 12 Hz

Answer: A

The graph shows 3 complete waves in 6 m, so the wavelength λ = 6/3 = 2 m. Frequency f = v/λ = 6/2 = 3 Hz.

7. The graph shows how the temperature of a block of ice, initially at −20 °C, changes with time when it is heated steadily. Which portion of the graph represents the boiling of the water?

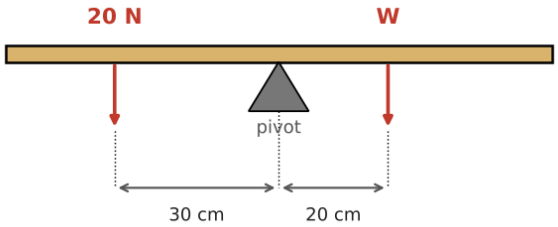


A. AB
B. BC
C. CD
D. DE

Answer: D

The flat portions are changes of state, where heat is absorbed without a rise in temperature. BC (0 °C) is melting; DE (100 °C) is boiling, when latent heat of vaporisation is absorbed.

8. A uniform metre rule is pivoted at its centre and balances horizontally with a 20 N weight and a weight W hung as shown. What is the value of W?

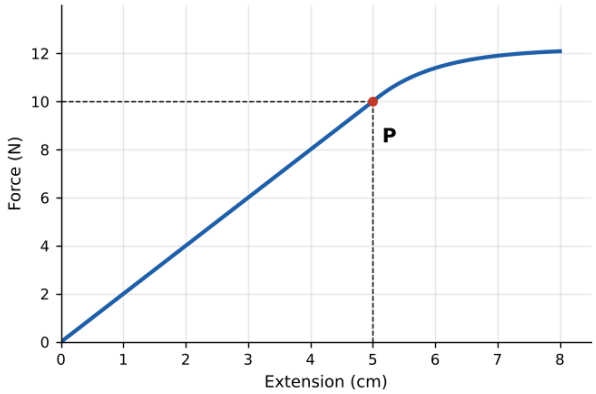


A. 13.3 N
B. 20 N
C. 30 N
D. 50 N

Answer: C

At balance, clockwise moment = anticlockwise moment: 20 × 30 = W × 20, so W = 600/20 = 30 N. (The rule's own weight acts at the pivot, so it has no moment.)

9. The graph shows how the extension of a spring varies with the force applied to it. P is the elastic limit. Calculate the force constant (spring constant) of the spring.

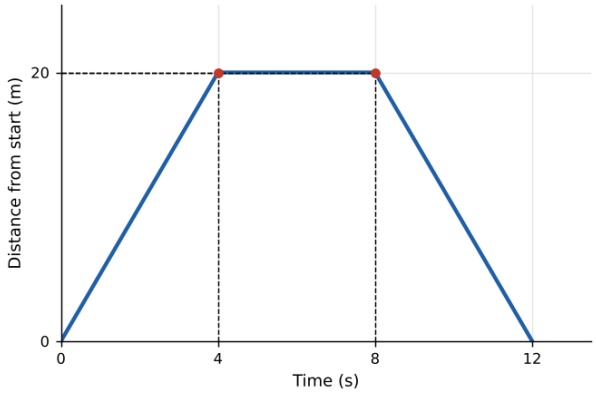


A. 200 N m^{-1}
B. 2 N m^{-1}
C. 50 N m^{-1}
D. 500 N m^{-1}

Answer: A

Within the elastic limit, k = F/e = 10 N / 0.05 m = 200 N m^{-1}. Note that 5 cm must be converted to 0.05 m.

10. The graph shows the distance of a cyclist from the starting point of a straight road at different times. What is the average speed of the cyclist for the whole 12 s?

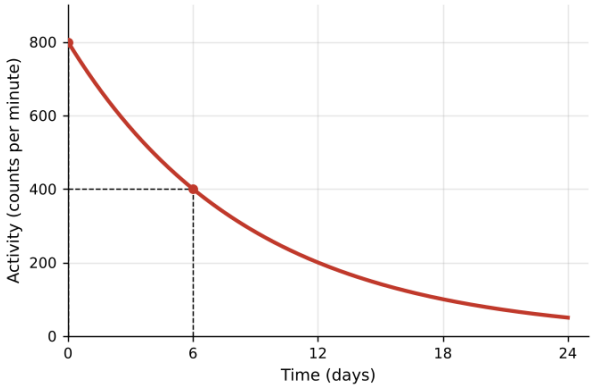


A. 5 m s^{-1}
B. 3.3 m s^{-1}
C. 0 m s^{-1}
D. 1.7 m s^{-1}

Answer: B

The cyclist goes 20 m away (4 s), rests (4 s), then returns 20 m (4 s). Total distance = 40 m in 12 s, so average speed = 40/12 ≈ 3.3 m s^{-1}. (Average velocity would be 0, since displacement is zero.)

11. The graph shows the decay of a radioactive sample. What will be the activity of the sample after 18 days?

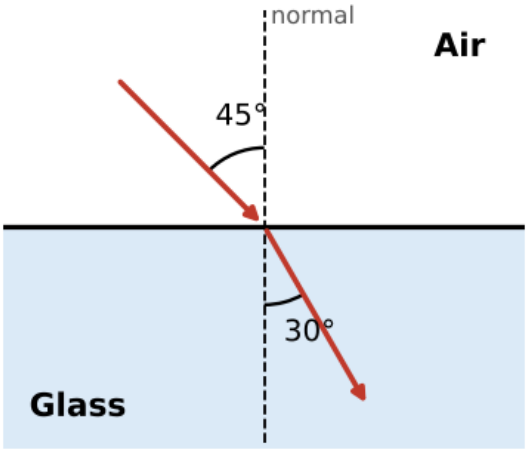


A. 50 counts per minute
B. 200 counts per minute
C. 400 counts per minute
D. 100 counts per minute

Answer: D

The activity falls from 800 to 400 in 6 days, so the half-life is 6 days. 18 days = 3 half-lives: 800 → 400 → 200 → 100 counts per minute.

12. A ray of light passes from air into glass as shown in the diagram. Calculate the refractive index of the glass.

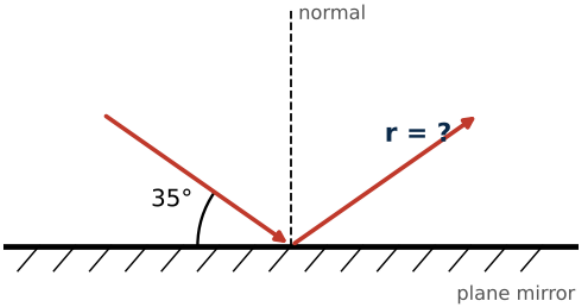


A. 1.22
B. 1.50
C. 1.41
D. 1.73

Answer: C

n = sin i / sin r = sin 45° / sin 30° = 0.707 / 0.5 = 1.41. (Angles are measured from the normal.)

13. A ray of light strikes a plane mirror so that it makes an angle of 35° with the surface of the mirror, as shown. What is the angle of reflection r?

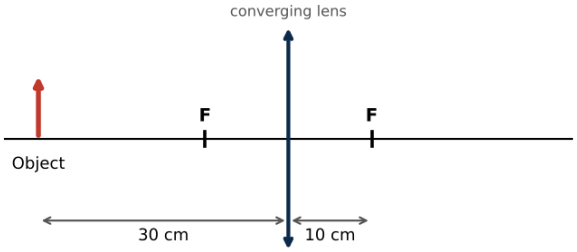


A. 55°
B. 35°
C. 70°
D. 110°

Answer: A

Angles of incidence and reflection are measured from the normal. Angle of incidence = 90° − 35° = 55°. By the law of reflection, r = 55°.

14. An object is placed 30 cm from a converging lens whose focal length is 10 cm, as shown. At what distance from the lens is the image formed?

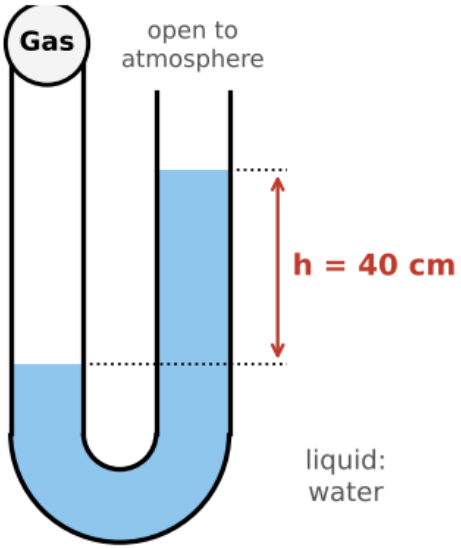


A. 7.5 cm
B. 20 cm
C. 15 cm
D. 30 cm

Answer: C

Lens formula: 1/f = 1/u + 1/v, so 1/v = 1/10 − 1/30 = 2/30 and v = 15 cm. The image is real and formed on the opposite side of the lens.

15. A U-tube manometer containing water is connected to a gas supply, as shown. By how much does the gas pressure exceed the atmospheric pressure? [density of water = 1000 kg m^{-3}, g = 10 m s^{-2}]

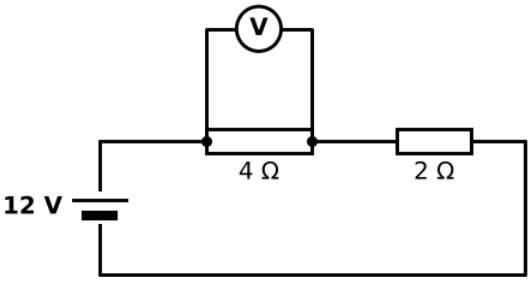


A. 400 Pa
B. 4 000 Pa
C. 40 000 Pa
D. 40 Pa

Answer: B

The gas pressure exceeds atmospheric pressure by hρg = 0.40 m × 1000 kg m^{-3} × 10 m s^{-2} = 4 000 Pa. (Convert 40 cm to 0.40 m.)

16. In the circuit shown, the battery has negligible internal resistance and the voltmeter V has a very high resistance. What is the reading of the voltmeter?

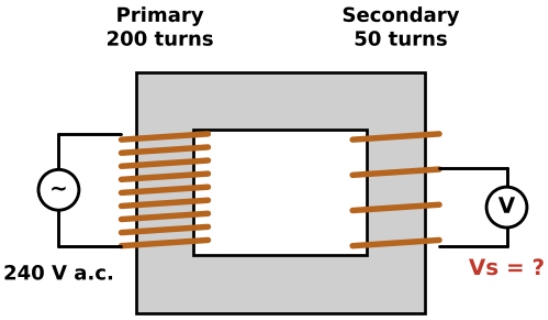


A. 4 V
B. 6 V
C. 12 V
D. 8 V

Answer: D

The resistors are in series: R = 4 + 2 = 6 Ω, so I = 12/6 = 2 A. The voltmeter is across the 4 Ω resistor, so V = IR = 2 × 4 = 8 V.

17. The diagram shows an ideal transformer. The primary coil has 200 turns and is connected to a 240 V a.c. supply, and the secondary coil has 50 turns. What is the secondary voltage Vs?

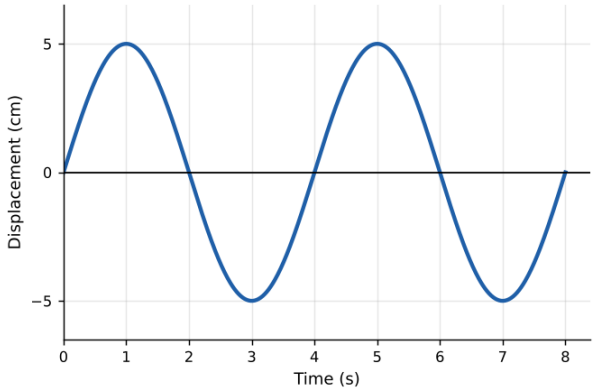


A. 60 V
B. 120 V
C. 48 V
D. 960 V

Answer: A

For an ideal transformer, Vs/Vp = Ns/Np. So Vs = 240 × 50/200 = 60 V. It is a step-down transformer because the secondary has fewer turns.

18. The graph shows the displacement of a simple pendulum bob with time as it oscillates. What is the frequency of the oscillation?

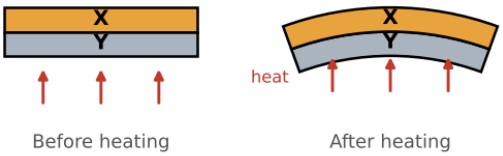


A. 0.50 Hz
B. 4.0 Hz
C. 0.25 Hz
D. 2.0 Hz

Answer: C

The graph shows 2 complete oscillations in 8 s, so the period T = 4 s. Frequency f = 1/T = 1/4 = 0.25 Hz.

19. A bimetallic strip made of two different metals X and Y is straight at room temperature. When heated, it bends as shown. Which of the following statements is correct?

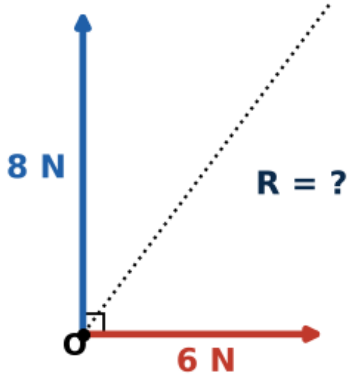


A. Y expands more than X.
B. X and Y have the same linear expansivity.
C. The strip bends towards the metal that expands more.
D. X expands more than Y.

Answer: D

The metal that expands more becomes longer and so must lie on the outer (convex) side of the curve. X is on the outside, so X has the greater linear expansivity. The strip bends towards the metal that expands less.

20. Two forces of 6 N and 8 N act at a point O at right angles to each other, as shown. What is the magnitude of their resultant?



A. 14 N
B. 10 N
C. 2 N
D. 48 N

Answer: B

The forces are perpendicular, so by Pythagoras R = √(6² + 8²) = √(36 + 64) = √100 = 10 N.